1. Ten major college football bowl games were played in January 2010, with the University of Alabama beating the University of Texas 37 to 21 to become the national champion of college football. The results of the 10 bowl games follow (*USA Today,* January 8, 2010). The predicted winning point margin was based on Las Vegas betting odds approximately one week before the bowl games were played. For example, Auburn was predicted to beat Northwestern in the Outback Bowl by five points. The actual winning point margin for Auburn was three points. A negative predicted winning point margin means that the team that won the bowl game was an underdog and expected to lose. For example, in the Rose Bowl, Ohio State was a two-point underdog to Oregon and ended up winning by nine points (**BowlGames dataset)**.

   a. Develop a scatter diagram with predicted point margin on the horizontal axis.
   b. What is the relationship between predicted and actual point margins?
   c. Compute and interpret the sample covariance.
   d. Compute the sample correlation coefficient. What does this value indicate about the relationship between the Las Vegas predicted point margin and the actual point margin in college football bowl games?

2. At the beginning of 2009, the economic downturn resulted in the loss of jobs and an increase in delinquent loans for housing. The national unemployment rate was 6.5% and the percentage of delinquent loans was 6.12% (The Wall Street Journal, January 27, 2009). In projecting where the real estate market was headed in the coming year, economists studied the relationship between the jobless rate and the percentage of delinquent loans. The expectation was that if the jobless rate continued to increase, there would also be an increase in the percentage of delinquent loans. The data below show the jobless rate and the delinquent loan percentage for 27 major real estate markets (**Housing dataset**).

   a. Compute the correlation coefficient. Is there a positive correlation between the jobless rate and the percentage of delinquent housing loans? What is your interpretation?
   b. Show a scatter diagram of the relationship between jobless rate and the percentage of delinquent housing loans.

3. The U.S. Census Bureau's American Community Survey reported the percentage of children under 18 years of age who had lived below the poverty level during the previous 12 months (U.S. Census Bureau website, August 2008). The region of the country, Northeast (NE), Southeast (SE), Midwest (MW), Southwest (SW), and West (W) and the percentage of children under 18 who had lived below the poverty level are shown for each state (**PovertyLevel dataset)**.

   a. What is the median poverty level percentage for the 50 states?
   b. What are the first and third quartiles? What is your interpretation of the quartiles?
   c. Show a box plot for the data. Interpret the box plot in terms of what it tells you about the level of poverty for children in the United States. Are any states considered outliers? Discuss.
   d. Identify the states in the lower quartile. What is your interpretation of this group and what region or regions are represented most in the lower quartile?